

# Ch5. Multiple Regression

## 5.1 The model

Multiple Linear Regression Model with  $p$  independent variables is defined as

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i \quad (5.1)$$

*observation*  
 $(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i=1, \dots, n$   
 $\epsilon_i$  or  $y_i$  are independent for  $i=1, \dots, n$

where  $i = 1, \dots, n$  ( $n$  is the sample size), or in matrix form  $\Sigma \sim (0, \sigma^2 I)$

$$\underline{y}_{n \times 1} = \underline{X}_{n \times r} \underline{\beta}_{r \times 1} + \underline{\epsilon}_{n \times 1}$$

where  $r = p + 1$ .

■ What are  $\underline{y}$ ,  $\underline{X}$ ,  $\underline{\beta}$  and  $\underline{\epsilon}$ ?

$\underline{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$       $\underline{X} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}$       $\underline{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$       $\underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$

$\downarrow$   $\underline{x}_i = \begin{pmatrix} x_{i1} \\ \vdots \\ x_{ip} \end{pmatrix}$       $\downarrow$   $\underline{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$       $\downarrow$   $\underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$

$r = p + 1$

$\leftarrow i$ -th subject  $\underline{x}_i' = (1, x_{i1}, \dots, x_{ip})$

■ Main assumptions of the model are

- 1  $E(\epsilon) = \mathbf{0}$ ;
- 2  $\text{Cov}(\epsilon) = \sigma^2 \mathbf{I}$ , thus  $\epsilon_i$  or  $y_i$  are uncorrelated for  $i = 1, \dots, n$ ;
- 3  $\underline{X}$  is full column rank (for now).  $\text{Rank}(\underline{X}) = p + 1$

## 5.2 Least Squares Estimation (LSE)

### 5.2.1 LSE

Minimize the sum of squares of deviations of observed and predicted values  $(\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$ , we have

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Remark 5.1

**Proof.**

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i \\ &= \tilde{x}_i' \beta + \varepsilon_i, \quad \varepsilon_i \text{ iid. } (0, \sigma^2) \end{aligned}$$

To find  $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$  and  $\hat{\sigma}^2$   
(sum of the squared errors)

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^n (y_i - \tilde{x}_i' \hat{\beta})^2$$

$$= (\underline{y} - \underline{x} \hat{\underline{\beta}})' (\underline{y} - \underline{x} \hat{\underline{\beta}})$$

$$SSE(\underline{\beta}) = (\underline{y} - \underline{x} \underline{\beta})' (\underline{y} - \underline{x} \underline{\beta})$$

$$= \underline{y}' \underline{y} - 2 \underline{y}' \underline{x} \underline{\beta} + \underline{\beta}' (\underline{x}' \underline{x}) \underline{\beta}$$

$$LSE \text{ of } \underline{\beta} = \arg \min_{\underline{\beta}} SSE(\underline{\beta})$$

$$\underline{0} = \frac{\partial SSE(\underline{\beta})}{\partial \underline{\beta}} = \underline{0} - 2 \underline{x}' \underline{y} + 2 (\underline{x}' \underline{x}) \underline{\beta}$$

$$\Rightarrow (\underline{x}' \underline{x}) \underline{\beta} = \underline{x}' \underline{y}$$

$$\text{If } \underline{x} \text{ is full rank} \Rightarrow \hat{\underline{\beta}} = (\underline{x}' \underline{x})^{-1} \underline{x}' \underline{y}$$

$$\underline{\hat{\sigma}^2} = ?$$

## 5.2.1 LSE – residuals and estimator of $\sigma^2$

- Vector of residuals  $\hat{\epsilon} = y - X\hat{\beta}$ :

$$\begin{aligned}\hat{\epsilon} &= y - X(X'X)^{-1}X'y \\ &= [I - X(X'X)^{-1}X']y \\ &= [I - H]y\end{aligned}$$

$$\hat{H}^2 = \hat{H}$$

where  $H = X(X'X)^{-1}X'$ .

- Fitted value:  $\hat{y} = X\hat{\beta} = X(X'X)^{-1}X'y = Hy$ .
- The unbiased estimator of  $\sigma^2$  is

$$\hat{\sigma}^2 = \frac{\hat{\epsilon}'\hat{\epsilon}}{n-p-1}$$

RSS

$$\hat{y} = X\hat{\beta} + \hat{\epsilon}$$

$$\begin{aligned}\hat{y} &= X\hat{\beta} \\ &= X(X'X)^{-1}X'y \\ &\hat{=} \hat{H}y\end{aligned}$$

$\hat{H}$  – hat matrix

$$\hat{H} = X(X'X)^{-1}X'$$



## 5.2.1 LSE – residuals and estimator of $\sigma^2$

- $\mathbf{X}'\hat{\epsilon} = \mathbf{0}$  ( $\mathbf{X}'\mathbf{H} = \mathbf{X}'$ ,  $\mathbf{H}\mathbf{X} = \mathbf{X}$  and  $\mathbf{X}'(\mathbf{I} - \mathbf{H}) = \mathbf{0}$ ,  $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$ ); ✓
- $\mathbf{H}$  is symmetric idempotent; ✓
- $\hat{\mathbf{y}}'\hat{\epsilon} = 0$ ; ✓
- $\mathbf{I} - \mathbf{H}$  is symmetric idempotent; ✓
- $E(\hat{\beta}) = \beta$  (unbiased estimator);
- $\text{Cov}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2$ ;
- $\text{tr}(\mathbf{I} - \mathbf{H}) = n - r$ ;
- $\hat{\epsilon}'\hat{\epsilon} = \text{tr}(\mathbf{y}\mathbf{y}'(\mathbf{I} - \mathbf{H}))$ ;
- $E(\mathbf{y}\mathbf{y}') = \sigma^2\mathbf{I} + \mathbf{X}\beta\beta'\mathbf{X}'$ ;
- $E\left(\frac{\hat{\epsilon}'\hat{\epsilon}}{n-r}\right) = \sigma^2$ .

Remark 5.2

Remark 5.2

$$\underset{\sim}{Y} = \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \text{cov}(\underset{\sim}{Y}) = \text{cov}(\underset{\sim}{\varepsilon}) = \sigma^2 \underset{\sim}{I}$$

$$E(\hat{\beta}) = E\left[(\underset{\sim}{X}'\underset{\sim}{X})^{-1}\underset{\sim}{X}'\underset{\sim}{Y}\right]$$

$$E\underset{\sim}{Y} = \underset{\sim}{X} \underset{\sim}{\beta}$$

$$= (\underset{\sim}{X}'\underset{\sim}{X})^{-1}\underset{\sim}{X}'(E\underset{\sim}{Y})$$

$$\underset{\sim}{Y} \sim (\underset{\sim}{X} \underset{\sim}{\beta}, \sigma^2 \underset{\sim}{I})$$

$$= (\underset{\sim}{X}'\underset{\sim}{X})^{-1}\underset{\sim}{X}'\underset{\sim}{X} \underset{\sim}{\beta} = \underset{\sim}{\beta}$$

$$SSE(\hat{\beta}) = (\underset{\sim}{Y} - \underset{\sim}{X} \hat{\beta})' (\underset{\sim}{Y} - \underset{\sim}{X} \hat{\beta})$$

$$\hat{\underset{\sim}{\varepsilon}} = \underset{\sim}{Y} - \underset{\sim}{X} \hat{\underset{\sim}{\beta}}$$

$$= \hat{\underset{\sim}{\varepsilon}}' \hat{\underset{\sim}{\varepsilon}}$$

$$= \underset{\sim}{Y}' (\underset{\sim}{I} - \underset{\sim}{H}) \underset{\sim}{Y}$$

$$E(\underset{\sim}{Y}' (\underset{\sim}{I} - \underset{\sim}{H}) \underset{\sim}{Y}) = \text{tr}((\underset{\sim}{I} - \underset{\sim}{H}) [\sigma^2 \underset{\sim}{I}]) + (\underset{\sim}{X} \underset{\sim}{\beta})' (\underset{\sim}{I} - \underset{\sim}{H}) \underset{\sim}{X} \underset{\sim}{\beta}$$

$$\begin{aligned} E(\underset{\sim}{Y}' \underset{\sim}{A} \underset{\sim}{Y}) &= \text{tr}(\underset{\sim}{A} \underset{\sim}{\Sigma}) \\ &+ \underset{\sim}{\mu}' \underset{\sim}{A} \underset{\sim}{\mu} \\ \underset{\sim}{Y} &\sim (\underset{\sim}{\mu}, \underset{\sim}{\Sigma}) \end{aligned}$$

$$= \sigma^2 \text{tr}(\underset{\sim}{I} - \underset{\sim}{H})$$

$$= \sigma^2 (n-r) = \sigma^2 (n-p-1)$$

$$(\underset{\sim}{I} - \underset{\sim}{H}) \underset{\sim}{X} = 0$$

$$\Rightarrow E\left(\frac{\hat{\underset{\sim}{\varepsilon}}' \hat{\underset{\sim}{\varepsilon}}}{n-r}\right) = \sigma^2$$

$\frac{\hat{\underset{\sim}{\varepsilon}}' \hat{\underset{\sim}{\varepsilon}}}{n-r}$  is an unbiased estimator of  $\sigma^2$ .

## 5.2.2 Generalized Least Squares Estimation

Assume that  $\text{Cov}(\epsilon) = \sigma^2 \mathbf{V}$ ,  $\mathbf{V}$  known

### ■ Estimation of $\beta$

- Ordinary Least Squares Estimator:

minimize  $(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \Rightarrow \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$

- Generalized Least Squares Estimator: minimize  $(\mathbf{y} - \mathbf{X}\mathbf{a})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{a})$

Let

$$\begin{aligned} S &= (\mathbf{y} - \mathbf{X}\mathbf{a})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{a}) \\ &= \mathbf{y}'\mathbf{V}^{-1}\mathbf{y} - 2\mathbf{y}'\mathbf{V}^{-1}\mathbf{X}\mathbf{a} + \mathbf{a}'\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\mathbf{a} \end{aligned}$$

$$\frac{\partial S}{\partial \mathbf{a}} = -2\mathbf{X}'\mathbf{V}^{-1}\mathbf{y} + 2\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\mathbf{a} = 0$$

WLS  
 $\min \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2$

$$\begin{aligned} \Rightarrow \mathbf{X}'\mathbf{V}^{-1}\mathbf{y} &= \mathbf{X}'\mathbf{V}^{-1}\mathbf{X}\mathbf{a} \\ \Rightarrow \tilde{\mathbf{a}} &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y} \end{aligned}$$

Note: If  $\mathbf{V} = \sigma^2 \mathbf{I}$ , OLS = GLS

- Weighted Least Squares Estimator.

$\tilde{\mathbf{y}} = \tilde{\mathbf{X}}\tilde{\beta} + \tilde{\epsilon}$   
 $\tilde{\epsilon} \sim (0, \sigma^2 \tilde{\mathbf{V}})$

$\tilde{\mathbf{V}} = \begin{pmatrix} v_{11} & \dots & v_{1n} \\ \vdots & & \vdots \\ v_{n1} & \dots & v_{nn} \end{pmatrix}$

$\tilde{\mathbf{V}}^{-1} = \begin{pmatrix} w_1 & & 0 \\ & \ddots & \\ 0 & & w_n \end{pmatrix}$

## 5.2.3 Properties of LSE

**Gauss-Markov Theorem:** The best linear unbiased estimator (b.l.u.e.): Let  $\mathbf{t}$  be a vector and we need to construct the b.l.u.e. of  $\mathbf{t}'\beta$ .

- Let  $\lambda'\mathbf{y}$  be a linear function of the observations and an estimator of  $\mathbf{t}'\beta$ .
- If  $\lambda'\mathbf{y}$  is an unbiased estimator of  $\mathbf{t}'\beta$ ,  $E(\lambda'\mathbf{y}) = \mathbf{t}'\beta$ .

$$\text{But } E(\lambda'\mathbf{y}) = \lambda'E(\mathbf{y}) = \lambda'\mathbf{X}\beta$$

Hence,  $\lambda'\mathbf{X}\beta = \mathbf{t}'\beta$  which is true for all  $\beta$

$$\Rightarrow \lambda'\mathbf{X} = \mathbf{t}'$$

- Find the linear unbiased estimator of  $\mathbf{t}'\beta$  which has minimum variance.

$$\text{Var}(\lambda'\mathbf{y}) = \sigma^2 \lambda' \mathbf{V} \lambda$$

Remark 3.3 GLSE is usually used in the case

that :  $\underline{y} = \underline{X} \underline{\beta} + \underline{\varepsilon}$  ,  $\underline{\varepsilon} \sim (0, \sigma^2 \underline{V})$

$$\begin{aligned} \underline{\tilde{\beta}} &= \underline{\tilde{\varepsilon}} = E \left[ (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \underline{X}' \underline{V}^{-1} \underline{y} \right] \\ &= (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \underline{X}' \underline{V}^{-1} \underline{X} \underline{\beta} \\ &= \underline{\beta} \end{aligned}$$

$$\begin{aligned} \text{Var}(\underline{\tilde{\varepsilon}}) &= (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \underline{X}' \underline{V}^{-1} \left[ \text{Var}(\underline{y}) \right] \\ &\quad \underline{V}^{-1} \underline{X} (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \end{aligned}$$

$$\begin{aligned} &= \sigma^2 (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \underline{X}' \underline{V}^{-1} \underline{V} \underline{V}^{-1} \underline{X} (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \\ &= \sigma^2 (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \end{aligned}$$

$$S^2 = \frac{SSE}{n-p-1} = (\underline{y} - \underline{X} \underline{\tilde{\varepsilon}})' \underline{V}^{-1} (\underline{y} - \underline{X} \underline{\tilde{\varepsilon}}) / (n-p-1) = \tilde{\sigma}^2$$

$$E \tilde{\sigma}^2 = E S^2 = \sigma^2$$

$$\underline{y} = \underline{X} \underline{\beta} + \underline{\varepsilon} \quad , \quad \underline{\varepsilon} \sim (0, \sigma^2 \underline{V})$$

$$\text{OLS: } \underline{\hat{\beta}} = (\underline{X}' \underline{X})^{-1} \underline{X}' \underline{y} \quad , \quad E(\underline{\hat{\beta}}) = \underline{\beta}$$

$$\begin{aligned} \text{Var}(\underline{\hat{\beta}}) &= (\underline{X}' \underline{X})^{-1} \underline{X}' \left[ \text{Var}(\underline{y}) \right] \underline{X} (\underline{X}' \underline{X})^{-1} \\ &= \tilde{\sigma}^2 (\underline{X}' \underline{X})^{-1} \underline{X}' \underline{V} \underline{X} (\underline{X}' \underline{X})^{-1} \\ &\geq \text{Var}(\underline{\tilde{\varepsilon}}) = \sigma^2 (\underline{X}' \underline{V}^{-1} \underline{X})^{-1} \end{aligned}$$

## 5.2.3 Properties of LSE

Gauss-Markov Theorem:  $W = \lambda' V \lambda$  is minimum if

$$\lambda' = t'(X' V^{-1} X)^{-1} X' V^{-1}$$

best  
linear  
unbiased  
estimator

subject to the constraint that  $X' \lambda = t$ ; i.e. GLS is the b.l.u.e.

Remark 5.4  $\underset{\sim}{y} = \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \underset{\sim}{\varepsilon} \sim (0, \sigma^2 \underset{\sim}{V}), \quad \underset{\sim}{V} \text{ given}$

$$\begin{aligned} \text{GLSE} \quad \underset{\sim}{\hat{\alpha}} &= (\underset{\sim}{X}' \underset{\sim}{V}^{-1} \underset{\sim}{X})^{-1} \underset{\sim}{X}' \underset{\sim}{V}^{-1} \underset{\sim}{y} \\ &\stackrel{d}{=} \underset{\sim}{A} \underset{\sim}{y} \end{aligned}$$

$$\boxed{\underset{\sim}{A} \stackrel{d}{=} (\underset{\sim}{X}' \underset{\sim}{V}^{-1} \underset{\sim}{X})^{-1} \underset{\sim}{X}' \underset{\sim}{V}^{-1}}$$

Gauss-Markov theorem:

$\underset{\sim}{t}' \underset{\sim}{\beta}$  | linear unbiased estimator.

$$\begin{aligned} E(\hat{\beta}) &= \beta \\ \Leftrightarrow \hat{X}' \hat{X} \beta &= \hat{X}' \beta \quad \text{for any } \beta \\ \Leftrightarrow \hat{X}' \hat{\beta} &= \hat{X}' \beta \end{aligned}$$

$$\begin{aligned} \hat{\beta} &= (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{Y} \\ &= \hat{X}'^{-1} \hat{X}' \hat{Y} \quad \hat{\beta} \text{ is a linear unbiased estimator of } \beta \end{aligned}$$

Need to prove  $\text{Var}(\hat{\beta}) = \min_{\hat{\beta} \in \{\hat{\beta} : E(\hat{\beta}) = \beta\}} \text{Var}(\hat{\beta})$

$$\text{Var}(\hat{\beta}) = \sigma^2 (\hat{X}' \hat{X})^{-1} \quad \text{OR in } \{\hat{\beta} : \hat{X}' \hat{\beta} = \hat{X}' \beta\}$$

$$\begin{aligned} (\hat{X}' \hat{X}) &= (\hat{X} - \hat{A}' \hat{\beta} + \hat{A}' \hat{\beta})' \hat{X} (\hat{X} - \hat{A}' \hat{\beta} + \hat{A}' \hat{\beta}) \\ &= (\hat{X} - \hat{A}' \hat{\beta})' \hat{X} (\hat{X} - \hat{A}' \hat{\beta}) + (\hat{A}' \hat{\beta})' \hat{X} (\hat{A}' \hat{\beta}) \\ &\quad + 2 (\hat{A}' \hat{\beta})' \hat{X} (\hat{X} - \hat{A}' \hat{\beta}) \end{aligned}$$

$$\begin{aligned} (\hat{A}' \hat{\beta})' \hat{X} (\hat{X} - \hat{A}' \hat{\beta}) &= \hat{\beta}' \hat{A}' \hat{X} \hat{X} - \hat{\beta}' \hat{A}' \hat{X} \hat{A}' \hat{\beta} \\ &= \hat{\beta}' (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{X} \hat{X} - \hat{\beta}' (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{X} \hat{X} (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{X} \hat{\beta} \\ &= \hat{\beta}' (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{X} - \hat{\beta}' (\hat{X}' \hat{X})^{-1} \hat{X}' \hat{X} = 0 \end{aligned}$$

$$\text{Var}(\hat{\beta}) = \sigma^2 (\hat{X}' \hat{X})^{-1} \geq \sigma^2 (\hat{A}' \hat{\beta})' \hat{X} (\hat{A}' \hat{\beta}) = \text{Var}(\hat{\beta})$$

$$\text{Var}(\hat{\beta}) = \text{Var}(\hat{\beta}) \Leftrightarrow \hat{\beta} = \hat{A}' \hat{\beta} \Leftrightarrow \hat{\beta}' \hat{X} = \hat{\beta}' \hat{X}$$

Note: For the model

$$\underset{\sim}{Y} = \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \underset{\sim}{\varepsilon} \sim (0, \sigma^2 \underset{\sim}{V})$$

OLSE:  $\hat{\beta}$   $\underset{\sim}{V}$  - given.

$$E(\hat{\beta}) = E[(X'X)^{-1} X'Y] = (X'X)^{-1} X'X\beta = \beta$$

—  $\hat{\beta}$  is LUE, but it is not the BLUE!

the BLUE is  $\hat{\alpha} = (X'V^{-1}X)^{-1} X'V^{-1}Y$

—  $\hat{\beta} = (X'X)^{-1} X'Y$  is the BLUE when  $\underset{\sim}{V} = \underset{\sim}{I}$ .

$$Y = (y_1, \dots, y_n)' \sim N(X\beta, \sigma^2 \underset{\sim}{V})$$

$$V = \begin{pmatrix} \frac{1}{m_1} & & 0 \\ & \frac{1}{m_2} & \\ 0 & & \ddots \\ & & & \frac{1}{m_n} \end{pmatrix}$$

$$\underline{X\beta = \beta_1}$$

weighted LSE (GLSE)

$$\min \sum_{i=1}^n m_i (y_i - \hat{y}_i)^2$$

$$\hat{\beta} = ? \quad \tilde{\beta} = ? \quad \text{BLUE?}$$

$$N(\beta_1, \sigma^2 \underset{\sim}{I}) \rightarrow N(\beta_1, \sigma^2 \underset{\sim}{V})$$

$$\text{WLSE } \tilde{\beta} = \frac{\sum m_i y_i}{\sum m_i} \quad \text{weighted average!}$$



## 5.3 Maximum likelihood estimation

A normal model is defined by (5.1) with an additional assumption

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon}_{n \times 1} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}), \quad \mathcal{D} = \{\mathbf{y}, \mathbf{X}\}$$

where  $\sigma^2$  is unknown.

- The likelihood function is

$$L(\boldsymbol{\beta}, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{n}{2}} e^{-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}$$

Take log, we have

$$\begin{aligned} \log L(\boldsymbol{\beta}, \sigma^2) &= -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}) \end{aligned}$$

## 5.3 Maximum likelihood estimation

- MLE for  $\beta$  and  $\sigma^2$

$$\frac{\partial \log L}{\partial \beta} = \frac{1}{\sigma^2} (\mathbf{X}'\mathbf{y} - \mathbf{X}'\mathbf{X}\beta)$$

$$\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

- Put the above 2 equations to zero and we obtain the MLEs:

$$\text{MLE}(\beta) = \hat{\beta} = \frac{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}}{\text{OLS}}$$

$$\text{MLE}(\sigma^2) = \hat{\sigma}^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \frac{1}{n} \text{SSE}$$

$$= \frac{1}{n} (\mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y})'(\mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y})$$

$$= \frac{1}{n} \mathbf{y}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{y} \quad \sim H$$

$$= \frac{1}{n} \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y} = \frac{1}{n} [\mathbf{y}'\mathbf{y} - \hat{\beta}'\mathbf{X}'\mathbf{y}]$$

## 5.3 Maximum likelihood estimation: properties

- Distribution of  $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \sim N(\beta, (\mathbf{X}'\mathbf{X})^{-1}\sigma^2)$  Since

$$\begin{aligned} E(\hat{\beta}) &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{y}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta = \beta \end{aligned}$$

$$\text{Cov}(\hat{\beta}) = \text{Cov}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}) = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2.$$

- Let  $SSE = \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y}$ :

- $\hat{\sigma}^2 = \frac{SSE}{n}$  is the MLE of  $\sigma^2$

- $\tilde{\sigma}^2 = \frac{SSE}{n-r(\mathbf{X})}$  is an unbiased estimator of  $\sigma^2$  with  $r(\mathbf{X}) = \text{rank of } \mathbf{X}$ .

- $\hat{\beta}$  and  $SSE$  are independent.

- Distribution of  $\hat{\sigma}^2$

$$\frac{[n - r(\mathbf{X})]\tilde{\sigma}^2}{\sigma^2} \sim \chi^2_{(n-r(\mathbf{X}))}.$$

Remark 5.5

$$Y \sim N(\underline{X}\beta, \sigma^2 \underline{I})$$

$$\hat{\beta} = (\underline{X}'\underline{X})^{-1} \underline{X}'Y \quad (\text{BLUE})$$

$$SSE = Y'(\underline{I} - H)Y$$

$$\text{since } (\underline{X}'\underline{X})^{-1} \underline{X}' (\sigma^2 \underline{I}) (\underline{I} - H) = 0$$

$$\Rightarrow \hat{\beta} \text{ and } SSE \text{ are independent}$$

Thm 4.2

$$\frac{SSE}{\sigma^2} = Y' \left[ \frac{\underline{I} - H}{\sigma^2} \right] Y$$

$$\left( \begin{array}{l} \sim \chi^2_{(n-r(\underline{X}), \lambda)} \\ \lambda = \frac{1}{2} (\underline{X}\beta)' \left[ \frac{\underline{I} - H}{\sigma^2} \right] (\underline{X}\beta) = 0 \\ \sim \chi^2_{n-r(\underline{X})} \end{array} \right.$$

$$\begin{aligned} \text{Rank}(\underline{I} - H) &= n - \text{rank}(\underline{X}) \\ &= n - r \quad \text{if } \underline{X} \text{ is} \\ &\quad r = p+1 \quad \text{column full rank} \end{aligned}$$

$$\begin{aligned} \text{Thm 4.1} \\ \frac{(\underline{I} - H)}{\sigma^2} \cdot [\sigma^2 \underline{I}] \\ &= \underline{I} - H \\ &\text{is idempotent} \end{aligned}$$

$$\Rightarrow \frac{n-r(X)}{\sigma^2} \cdot \hat{\sigma}^2$$

$$= \frac{SSE}{\sigma^2} \sim \chi^2_{n-r(X)}$$

$$\hat{\sigma}^2 = \frac{SSE}{n-r(X)}$$

$$E(\hat{\sigma}^2) = \frac{\sigma^2 E(SSE/\sigma^2)}{(n-r(\underline{X}))} = \frac{n-r(\underline{X})}{n-r(\underline{X})} \sigma^2 = \sigma^2$$

$$\text{MLE of } \sigma^2 = \hat{\sigma}^2 = \frac{SSE}{n}, \quad E\hat{\sigma}^2 = \frac{n-r(\underline{X})}{n} \sigma^2 \rightarrow \sigma^2 \text{ as } n \rightarrow \infty$$

## 5.3 Maximum likelihood estimation: Example 5.1

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i = 1, 2, \dots, n; \quad \varepsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

$\text{ch2}$        $\underline{p=1}$        $\gamma = 1+1=2$

In matrix notation,  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}$$

$$\begin{aligned} (\mathbf{X}'\mathbf{X})^{-1} &= \frac{1}{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \begin{bmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{bmatrix} \\ &= \frac{1}{n \sum_{i=1}^n (x_i - \bar{x})^2} \begin{bmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{bmatrix} \end{aligned}$$

## 5.3 Maximum likelihood estimation: Example 5.1

$$\begin{aligned}
 \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
 &= \frac{1}{n \sum_{i=1}^n (x_i - \bar{x})^2} \begin{bmatrix} \sum_{i=1}^n y_i \sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i \sum_{i=1}^n x_i y_i \\ n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i \end{bmatrix} \\
 &= \begin{bmatrix} \bar{y} - \hat{\beta}_1 \bar{x} \\ \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{bmatrix} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix}
 \end{aligned}$$

$\frac{s_{xy}}{s_{xx}}$

## 5.3 Maximum likelihood estimation: Example 5.1

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n-2}(\mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}) \\&= \frac{1}{n-2}\left[\sum_{i=1}^n y_i^2 - \hat{\beta}_0 \sum_{i=1}^n y_i - \hat{\beta}_1 \sum_{i=1}^n x_i y_i\right] \\&= \frac{1}{n-2}\left[\sum_{i=1}^n y_i^2 - \bar{y} \sum_{i=1}^n y_i + \hat{\beta}_1 \bar{x} \sum_{i=1}^n y_i - \hat{\beta}_1 \sum_{i=1}^n x_i y_i\right] \\&= \frac{1}{n-2}\left[\sum_{i=1}^n (y_i - \bar{y})^2 - \hat{\beta}_1 \left(\sum_{i=1}^n x_i y_i - n \sum_{i=1}^n x_i \sum_{i=1}^n y_i\right)\right] \\&= \frac{1}{n-2}\left[\sum_{i=1}^n (y_i - \bar{y})^2 - \frac{[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right]\end{aligned}$$

## 5.4. The model in centered form

- The centered form of the multiple linear regression model is

$$\begin{aligned} y_i &= \underline{\beta_0} + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i \\ &= \alpha + \underline{b_1(x_{i1} - \bar{x}_1)} + b_2(x_{i2} - \bar{x}_2) + \cdots + b_p(x_{ip} - \bar{x}_p) + \epsilon_i \end{aligned}$$

where  $\alpha = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \cdots + \beta_p \bar{x}_p$ , and  $\bar{x}_j$  is the average of  $\{x_{ij}, i = 1, \dots, n\}$  for  $j = 1, 2, \dots, p$ .

- The matrix form can be expressed as

$$\mathbf{y}_{n \times 1} = \begin{pmatrix} \mathbf{1} & \mathbf{Z} \end{pmatrix} \begin{pmatrix} \alpha \\ \mathbf{b} \end{pmatrix} + \boldsymbol{\epsilon}_{n \times 1},$$

where  $(\mathbf{Z} = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{X}}')$ ,  $\bar{\mathbf{X}}' = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_p)$  and  $\mathbf{b}' = (b_1, \dots, b_p)$ .

- We can prove that

$$\hat{\alpha} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \hat{\beta}_2 \bar{x}_2 + \cdots + \hat{\beta}_p \bar{x}_p$$

and  $\hat{\mathbf{b}} = \hat{\boldsymbol{\beta}}_1$ .



Remark 5.6 (The model in centered form)

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i \quad (*)$$

$$\Rightarrow \hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$$

2<sup>nd</sup> centered form:

$$y_i = \alpha + \beta_1 (x_{i1} - \bar{x}_1) + \dots + \beta_p (x_{ip} - \bar{x}_p) + \varepsilon_i$$

$$\stackrel{d}{=} \alpha + b_1 z_{i1} + \dots + b_p z_{ip} + \varepsilon_i$$

$$\Rightarrow \hat{\alpha}, \hat{b}_1, \dots, \hat{b}_p$$

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

$$\alpha = \beta_0 + \beta_1 \bar{x}_1 + \dots + \beta_p \bar{x}_p$$

From (\*), 
$$\underset{n \times (p+1)}{\tilde{X}} = \begin{bmatrix} 1 & x_{i1} & \dots & x_{ip} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} = \begin{bmatrix} 1 & \tilde{X}_1 \end{bmatrix}$$

$\underbrace{\tilde{X}_1}_{n \times p}$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{pmatrix} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \sim \end{pmatrix} = (\tilde{X}' \tilde{X})^{-1} \tilde{X}' \tilde{y}$$

$$\underset{p \times 1}{\tilde{\beta}} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

$$= \left[ \begin{pmatrix} 1' \\ \tilde{X}_1' \end{pmatrix} (1 \ \tilde{X}_1) \right]^{-1} \begin{pmatrix} 1' \\ \tilde{X}_1' \end{pmatrix} \tilde{y} \quad \tilde{\tilde{X}} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{pmatrix}$$

$$= \begin{bmatrix} n & n \bar{\tilde{X}}' \\ n \bar{\tilde{X}} & \tilde{X}_1' \tilde{X}_1 \end{bmatrix}^{-1} \begin{pmatrix} n \bar{\tilde{y}} \\ \tilde{X}_1' \tilde{y} \end{pmatrix}$$

$$\begin{pmatrix} n & n \bar{\tilde{X}}' \\ n \bar{\tilde{X}} & \tilde{X}_1' \tilde{X}_1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{n} + \bar{\tilde{X}}' B^{-1} \bar{\tilde{X}} & -\bar{\tilde{X}}' B^{-1} \\ -B^{-1} \bar{\tilde{X}} & B^{-1} \end{pmatrix}$$

$$\begin{cases} \underline{A} = \begin{pmatrix} \underline{A}_{11} & \underline{A}_{12} \\ \underline{A}_{21} & \underline{A}_{22} \end{pmatrix} \\ \underline{A}^{-1} = \begin{pmatrix} & \\ & \end{pmatrix} \\ \underline{B} = \underline{X}_1' \underline{X}_1 - n \bar{\tilde{X}} \bar{\tilde{X}}' \end{cases}$$

$$\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} \frac{1}{n} + \bar{\underline{x}}' \underline{B}^{-1} \bar{\underline{x}} & -\bar{\underline{x}}' \underline{B}^{-1} \\ -\underline{B}^{-1} \bar{\underline{x}} & \underline{B}^{-1} \end{pmatrix} \begin{pmatrix} n\bar{y} \\ \underline{x}', \underline{y} \end{pmatrix}$$

$$= \begin{pmatrix} \bar{y} - \bar{\underline{x}}' \underline{B}^{-1} (\underline{x}', \underline{y} - n\bar{y} \bar{\underline{x}}) \\ \underline{B}^{-1} (\underline{x}', \underline{y} - n\bar{y} \bar{\underline{x}}) \end{pmatrix} \quad \hat{\beta}_1 \quad (***)$$

$$\Rightarrow \hat{\beta}_1 = \underline{B}^{-1} (\underline{x}', \underline{y} - n\bar{y} \bar{\underline{x}}), \quad \hat{\beta}_0 = \bar{y} - \bar{\underline{x}}' \hat{\beta}_1$$

From (\*\*),  $\underline{y} = (\underline{1}' \underline{z}) \begin{pmatrix} \underline{\alpha} \\ \underline{b} \end{pmatrix} + \underline{\varepsilon}$   $\underline{z} = (\underline{x}_1, -\underline{1} \bar{\underline{x}}')$   
 $\underline{\bar{z}} = 0$

$$\underline{B}^* = \underline{z}' \underline{z} = (\underline{x}_1, -\underline{1} \bar{\underline{x}}')' (\underline{x}_1, -\underline{1} \bar{\underline{x}}') = \underline{B}$$

Similar to (\*\*\*), we have

$$\begin{pmatrix} \hat{\underline{\alpha}} \\ \hat{\underline{b}} \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \underline{B}^* (\underline{z}', \underline{y}) \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \underline{B} (\underline{x}_1, -\underline{1} \bar{\underline{x}}')' \underline{y} \end{pmatrix}$$

$$= \begin{pmatrix} \bar{y} \\ \underline{B} (\underline{x}_1, \underline{y} - n\bar{y} \bar{\underline{x}}) \end{pmatrix} = \begin{pmatrix} \hat{\underline{\alpha}} \\ \hat{\beta}_1 \end{pmatrix}$$

$$\hat{\underline{b}} = \hat{\beta}_1, \quad \hat{\beta}_0 = \bar{y} - \bar{\underline{x}}' \hat{\beta}_1$$

$$= \hat{\underline{\alpha}} - \bar{\underline{x}}' \hat{\beta}_1$$

OR.  $\hat{\alpha} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \dots + \hat{\beta}_p \bar{x}_p$  ✗

## 5.4. The model in centered form

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \text{Var} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\mathbf{b}} \end{pmatrix} = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2 \\ &= \begin{bmatrix} \frac{1}{n} + \bar{\mathbf{X}}'(\mathbf{Z}'\mathbf{Z})^{-1}\bar{\mathbf{X}} & -\bar{\mathbf{X}}'(\mathbf{Z}'\mathbf{Z})^{-1} \\ -(\mathbf{Z}'\mathbf{Z})^{-1}\bar{\mathbf{X}} & (\mathbf{Z}'\mathbf{Z})^{-1} \end{bmatrix} \sigma^2\end{aligned}$$

Then

$$\begin{aligned}\text{Var}(\hat{\mathbf{b}}) &= (\mathbf{Z}'\mathbf{Z})^{-1}\sigma^2 \\ \text{Var}(\hat{\beta}_0) &= \frac{\sigma^2}{n} + \bar{\mathbf{X}}'(\mathbf{Z}'\mathbf{Z})^{-1}\bar{\mathbf{X}}\sigma^2 \\ &= \frac{\sigma^2}{n} + \bar{\mathbf{X}}' \text{Var}(\hat{\mathbf{b}}) \bar{\mathbf{X}} \\ \text{Cov}(\hat{\beta}_0, \hat{\mathbf{b}}') &= -\bar{\mathbf{X}}'(\mathbf{Z}'\mathbf{Z})^{-1}\sigma^2 = -\bar{\mathbf{X}}' \text{Var}(\hat{\mathbf{b}})\end{aligned}$$

## 5.5. Partitioning Total Sum of Squares

- SST (Total sum of squares corrected for the mean),

$$SST = \mathbf{y}'\mathbf{y} - \frac{1}{n}\mathbf{y}'\mathbf{1}\mathbf{1}'\mathbf{y} = \mathbf{y}'(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}')\mathbf{y}$$

$$\frac{SST}{\sigma^2} \sim \chi^2_{(n-1, \frac{\beta'\mathbf{X}'\mathbf{X}\beta - \frac{1}{n}(\mathbf{1}'\mathbf{X}\beta)^2}{2\sigma^2})}$$

- SSR (Sum of squares of regression ) = SST - SSE,

$$SSR = \mathbf{y}'(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}')\mathbf{y} - \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y}$$

$$= \mathbf{y}'(\mathbf{H} - \frac{1}{n}\mathbf{1}\mathbf{1}')\mathbf{y}$$

$$\frac{SSR}{\sigma^2} \sim \chi^2_{(k, \frac{\mathbf{b}'(\mathbf{Z}'\mathbf{Z})\mathbf{b}}{2\sigma^2})}$$

$$- R^2 = \frac{SSR}{SST}.$$

## 5.5. Partitioning Total Sum of Squares: Anova Table

| Source     | df                  | SS                                                                         | MS                              | F-statistics          |
|------------|---------------------|----------------------------------------------------------------------------|---------------------------------|-----------------------|
| Regression | $r(\mathbf{X}) - 1$ | $\hat{\mathbf{b}}' \mathbf{Z}' \mathbf{y}$                                 | $\frac{SSR}{r(\mathbf{X}) - 1}$ | $F = \frac{MSR}{MSE}$ |
| Error      | $n - r(\mathbf{X})$ | $\mathbf{y}' \mathbf{y} - \hat{\beta}' \mathbf{X}' \mathbf{y}$             | $\frac{SSE}{n - r(\mathbf{X})}$ |                       |
| Total      | $n - 1$             | $\mathbf{y}' (\mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}') \mathbf{y}$ |                                 |                       |

- SSR is independent of SSE
- $F \sim F_{(r(\mathbf{X})-1, n-r(\mathbf{X}), \frac{\mathbf{b}'(\mathbf{Z}'\mathbf{Z})\mathbf{b}}{2\sigma^2})}$
- Under  $H_0 : \mathbf{b} = \mathbf{0}$ ,

$$F \sim F_{(r(\mathbf{X})-1, n-r(\mathbf{X}), 0)}.$$

## 5.6. Model Misspecification – Misspecification of the error structure

- Suppose the true model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \text{Cov}(\mathbf{y}) = \sigma^2 \mathbf{V},$$

- but the working model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \text{Cov}(\mathbf{y}) = \sigma^2 \mathbf{I}.$$

- This will still have an unbiased estimate of  $\boldsymbol{\beta}$ , but it is not the BLUE.

## 5.6. Model Misspecification – Misspecification of the mean

We consider the following two models

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\epsilon}, \quad \text{Cov}(\mathbf{y}) = \sigma^2 \mathbf{I}, \quad (5.2)$$

and

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon}, \quad \text{Cov}(\mathbf{y}) = \sigma^2 \mathbf{I}. \quad (5.3)$$

- Under-fitting: if the true model is (5.2), but use the model (5.3);
- Over-fitting: if the true model is (5.3), but use the model (5.2).